

Jeff Shen

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Education

H.B.Sc., Statistics, Astronomy & Astrophysics, Mathematics
University of Toronto

2018-2022

Technical Skills

Data: Python, NumPy, SciPy, Pandas, PyTorch, Scikit-Learn, Matplotlib, Stan, SQL, NoSQL, R
Other: Rust, BLAS/LAPACK, Docker, bash, vim, git, AWS, SymPy, Astropy, NLTK, L^AT_EX, regex

Research Experience

- 2021 May - 2021 Aug. **Undergraduate Summer Research Fellowship**
Canadian Institute for Theoretical Astrophysics
Stuff here.
- 2020 Aug. - 2021 May **Undergraduate Research Course**
Canadian Institute for Theoretical Astrophysics
Supervised by [Prof. Gwendolyn Eadie](#) and [Prof. Norman Murray](#). Study of mass distribution of the Milky Way in a Bayesian framework. Implemented a hierarchical Bayesian model for modelling the mass of galactic halos using an adaptive gradient-based sampling algorithm, increasing computational efficiency by up to two orders of magnitude over existing code.
- 2020 May - 2020 Aug. **Summer Undergraduate Research Program**
Dunlap Institute for Astronomy and Astrophysics, University of Toronto
Supervised by [Prof. Allison Man](#). Study of molecular gas in a strongly-lensed galaxy group at $z = 2.9$. Awarded best poster in student competition. Analyzed ALMA data to infer molecular gas masses in several high-redshift galaxies. Resulted in a first-author publication.

Publications

- 2021 May **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. “Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$ ”, submitted to *ApJ*

Honours and Awards

- 2021 May - 2021 Aug. **Undergraduate Summer Research Fellowship, (\$8396 CAD)**
Canadian Institute for Theoretical Astrophysics, University of Toronto
- 2020 May - 2020 Aug. **Summer Undergraduate Research Program Award (\$9,500 CAD)**
Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Presentations

- 2021 May **Poster Presentation**
Canadian Astronomical Society (CASCA) AGM, Virtual

Outreach

- 2019 - Present **Space Science Journalist (Current in Space segment)**, *The Star Spot* podcast
Curate, write and record stories about the latest research and news in astronomy. Episodes 177 - present.

Interests

Languages: English (native), French, Chinese, Norwegian (basic)

Other: *nix customization, computer graphics, 3D modelling, running, baking, reading